topcatswap

User guide: web app (computer, phone, tablet) and desktop app

Uniswap V2, V3 and V4 swaps · Ethereum, Arbitrum and BSC

Ledger (USB / Bluetooth) · Trezor · MetaMask · OKX · Rabby · Coinbase Wallet · Trust Wallet · imported keys

Web 1.1.3 · Desktop 1.1.2 · English (US)

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1 What TopCatSwap is

TopCatSwap compares the Uniswap V2, V3 and V4 pools for a token pair, quotes each one with your real amount and swaps on the one that pays the most. It protects you against sandwich and frontrun attacks, and every transaction is signed by your own wallet. There is no TopCatSwap server: the code talks straight to the blockchain and to your wallet, and it is open source.

Way to use it	Where	Best for
Web app	https://topcatmasterinvestor.github.io/topcatswap/ (computer, phone or tablet)	Using it right away, on any device.
Ubuntu desktop app	Built from the code (see the Installation guide)	Daily use with a Ledger/Trezor over USB; bundles without a relay.
Command line	node desktop/engine/topcatswap.en.js	Scripts and advanced users.

- Networks: Ethereum, Arbitrum One and BNB Smart Chain.
- Wallets: Ledger, Trezor, MetaMask, OKX Wallet, Rabby, Coinbase Wallet, Trust Wallet and imported private keys (up to 5, each encrypted with its own password).
- Bundles: several transactions in a single block, sent privately to block builders, all-or-nothing, and no gas if it fails.
- English by default; Português with the PT | EN switch.

2 Opening the web app

Open https://topcatmasterinvestor.github.io/topcatswap/ in your browser. The first time it opens in English; tap PT for Portuguese. Your choice, settings and imported wallets are saved only in this browser, on this site.

The site adapts to the device. On a computer you get the sidebar layout. On a phone or tablet the wallet bar is at the top, the navigation at the bottom and everything is in one column.

Device and browser	What works
Computer: Chrome, Edge, Brave, Opera	Everything: Ledger over USB, Trezor, browser extensions, imported wallets.
Computer: Firefox	Trezor, extensions and imported wallets. Ledger shows as unavailable (Firefox has no WebHID).
Computer: Safari	Extensions and imported wallets. Ledger and Trezor show as unavailable.
Android: Chrome, Edge, Brave	Ledger Nano X, Stax or Flex over Bluetooth (experimental), wallet apps, imported wallets.
iPhone / iPad (any browser)	Wallet apps and imported wallets. Hardware wallets are hidden: iOS gives websites no USB or Bluetooth.
Inside a wallet app's browser	That wallet connects directly (MetaMask, Trust, OKX, Coinbase Wallet).

3 Choosing your wallet

Each time the site opens, it shows the wallet menu with the wallets that make sense on your device. The last one used is preselected. To change later, use “Change wallet” in the wallet card.

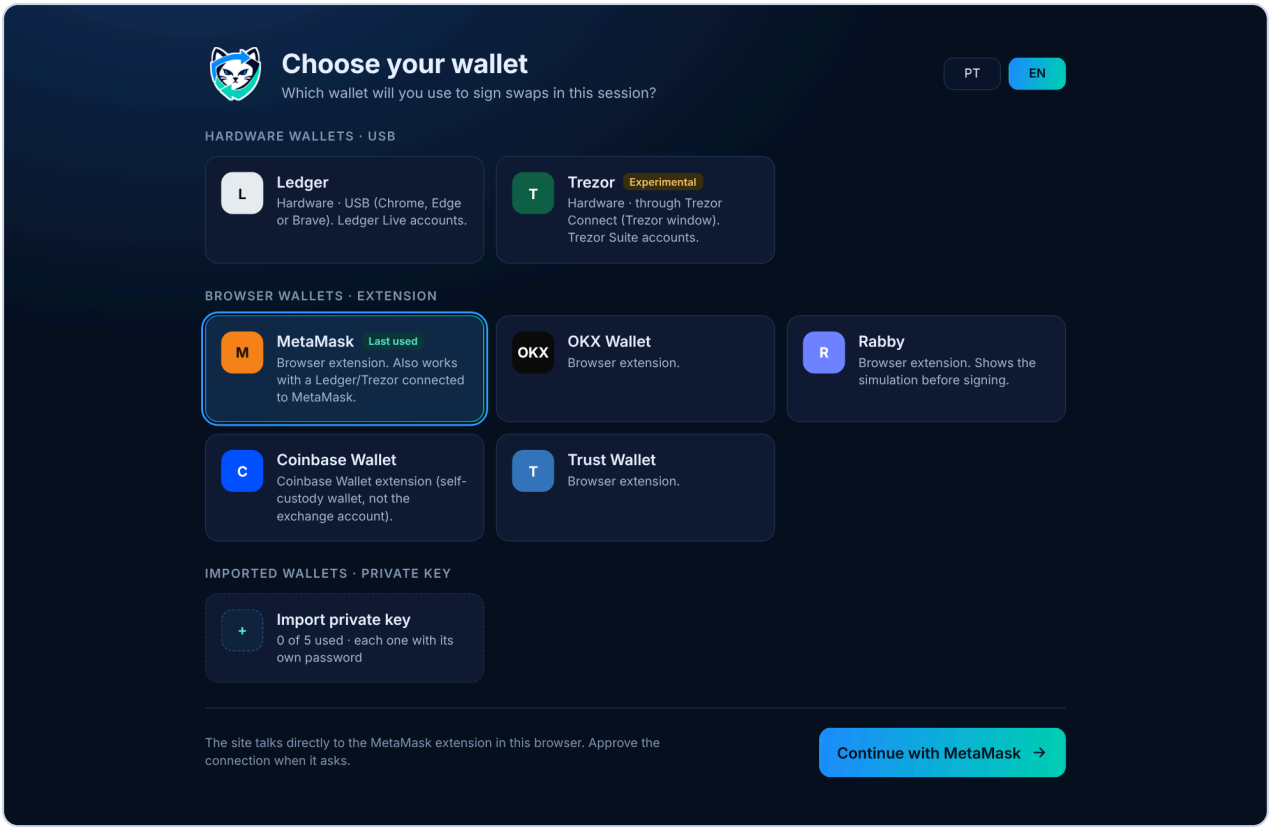


Figure 1. Wallet menu on a computer (Chrome).

Type	How it signs
Ledger / Trezor	On the device's screen. The key never leaves it.
Browser extension or wallet app	In the wallet's own window. The key stays in the wallet.
Imported wallet	In the page, after you type that wallet's password. The key is encrypted in this browser.

4 On a phone or tablet

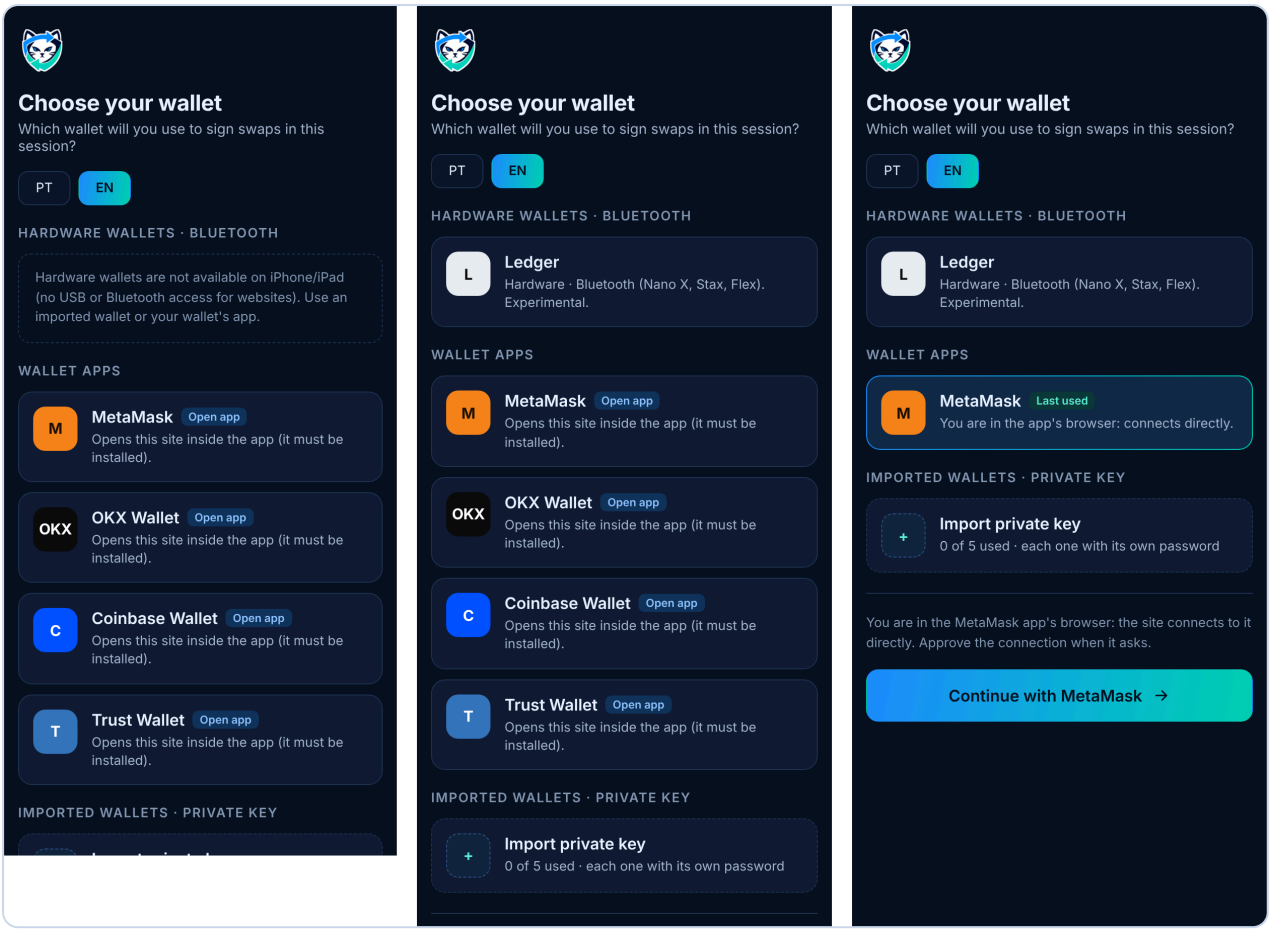


Figure 2. Wallet menu on iPhone (no hardware wallets), on Android (Ledger over Bluetooth) and inside the MetaMask app.

MetaMask, Trust, OKX, Coinbase Wallet

Phone browsers have no wallet extensions, so each wallet has an “Open app” button. It opens this site inside that wallet's own browser. There the wallet is already connected to the page: choose it, tap Continue and approve the connection. You need the wallet app installed.

Imported wallet

Works on every phone and tablet. The key is encrypted in the phone's browser with the password you choose. Opening it takes a few seconds on a phone (strong encryption).

Ledger over Bluetooth (Android, experimental)

1. Use Chrome, Edge or Brave on Android. A Ledger Nano X, Stax or Flex is required (the Nano S has no Bluetooth).
2. On the Ledger: unlock it, open the Ethereum app and turn Bluetooth on. In the Ethereum app settings, turn on “Blind signing”.
3. Choose Ledger in the menu and tap Continue. The browser asks you to pick the Ledger; the phone may ask for a pairing code shown on the Ledger.
4. If the connection drops, tap “Connect Ledger” in the top bar.

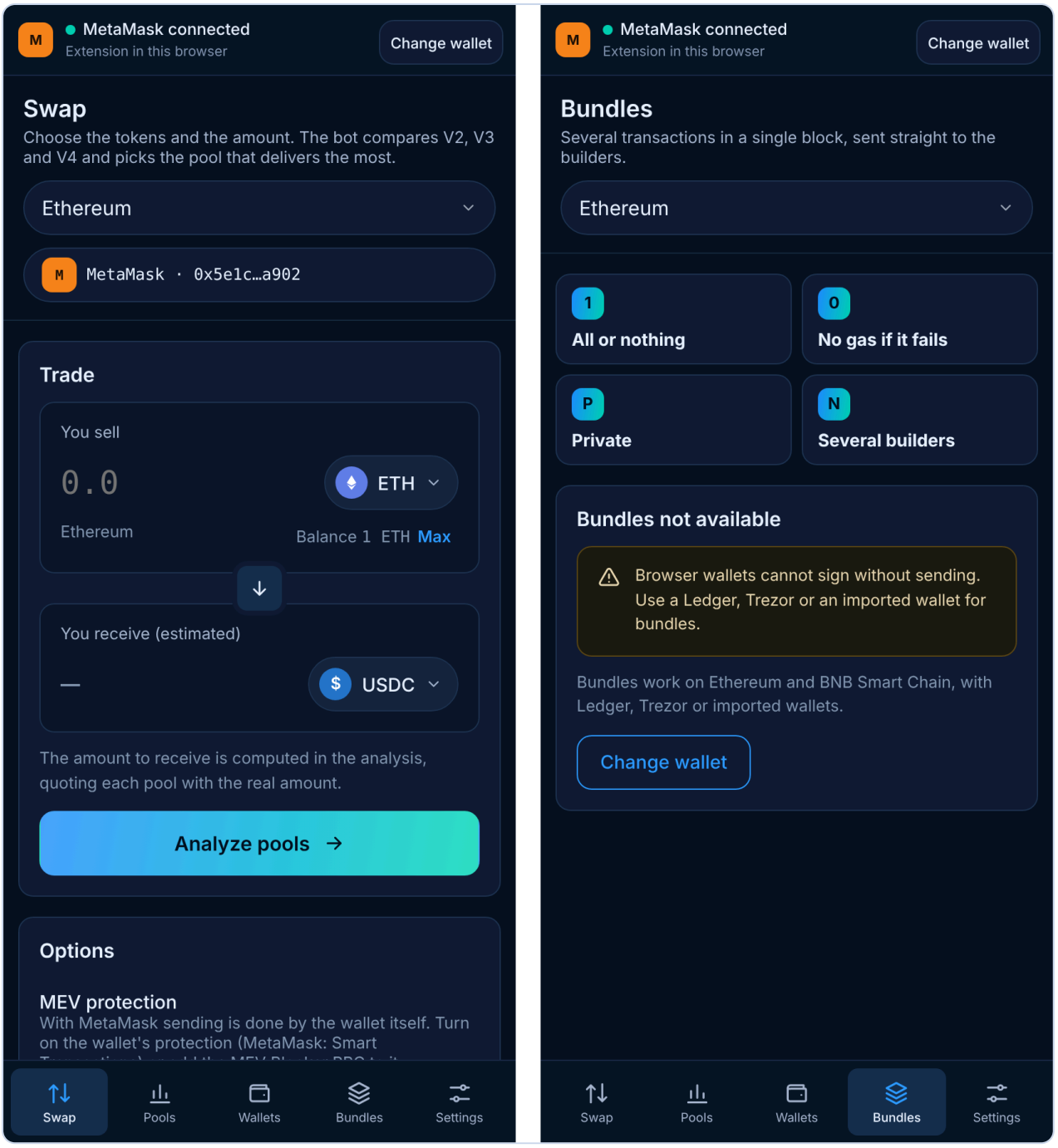


Figure 3. Swap and Bundles on a phone.

> Bundles on a phone need a wallet that signs without sending: an imported wallet or a Ledger over Bluetooth. Wallet apps (MetaMask & co.) cannot sign without sending.

5 Ledger and Trezor on a computer

Ledger over USB (Chrome, Edge, Brave, Opera)

- Close Ledger Live: only one app at a time can talk to the Ledger.
- Unlock the Ledger, open the Ethereum app and turn on “Blind signing” in its settings.
- The first time, the browser shows a window to pick the Ledger. Afterwards it is remembered. If it disappears, click “Authorize Ledger” in the wallet card.
- On Linux, install the Ledger udev rules once (see the Installation guide).

Trezor

- Uses the official Trezor Connect, which opens its own window (connect.trezor.io) for the PIN, passphrase and confirmation.
- Close Trezor Suite first. Not available in Safari or on phones.
- Accounts m/44'/60'/0'/0/N, the same as Trezor Suite and MetaMask.

6 Browser extensions on a computer

MetaMask, OKX Wallet, Rabby, Coinbase Wallet and Trust Wallet are found automatically in the browser (EIP-6963). Choose the wallet and click “Connect”; approve in the extension. Each approve and swap appears in the extension's window.

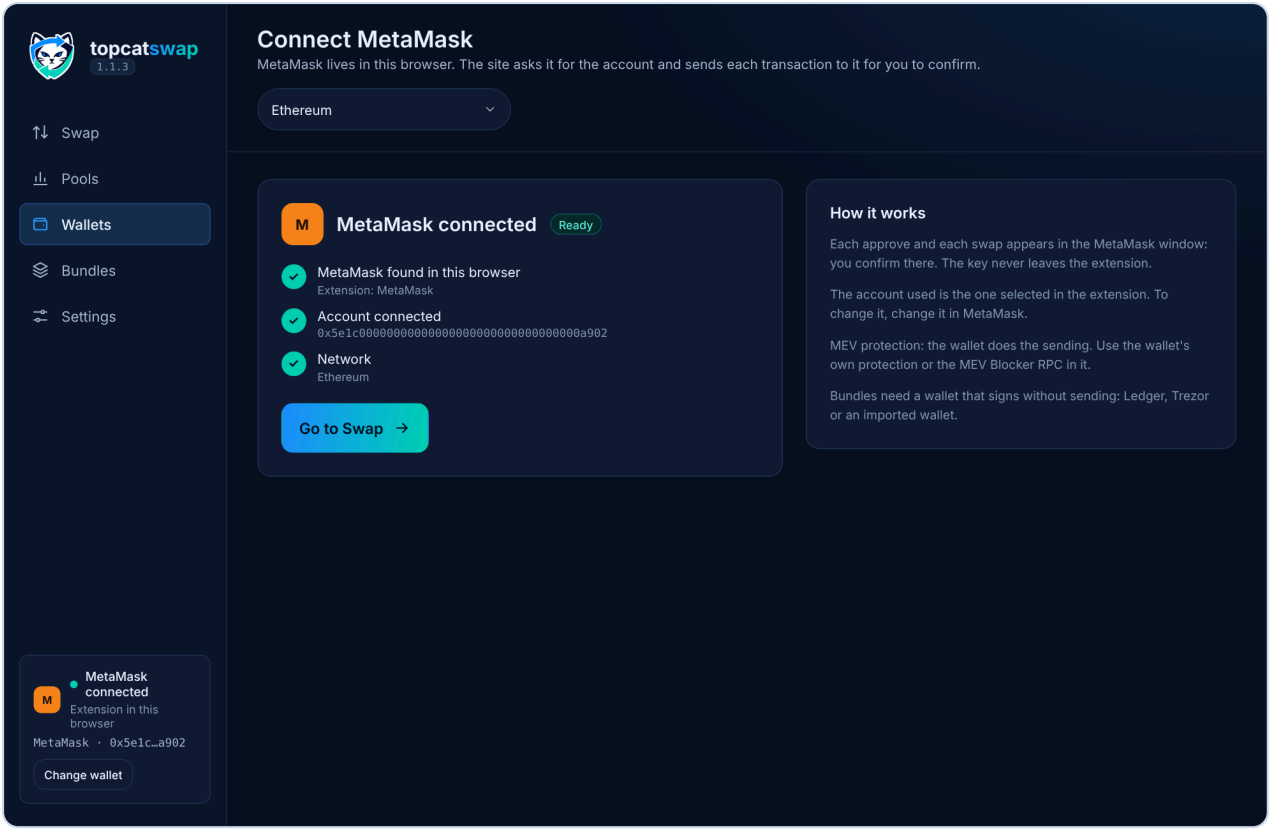


Figure 4. MetaMask connected on the web app.

> With these wallets the wallet itself sends the transaction. Use its own MEV protection, or set the MEV Blocker RPC in it.

7 Imported wallets

You can import up to 5 wallets by private key. Each one is encrypted with its own password (standard Ethereum keystore, scrypt + AES), on the web in this browser's storage and on the desktop app in ~/.config/TopCatSwap/wallets.

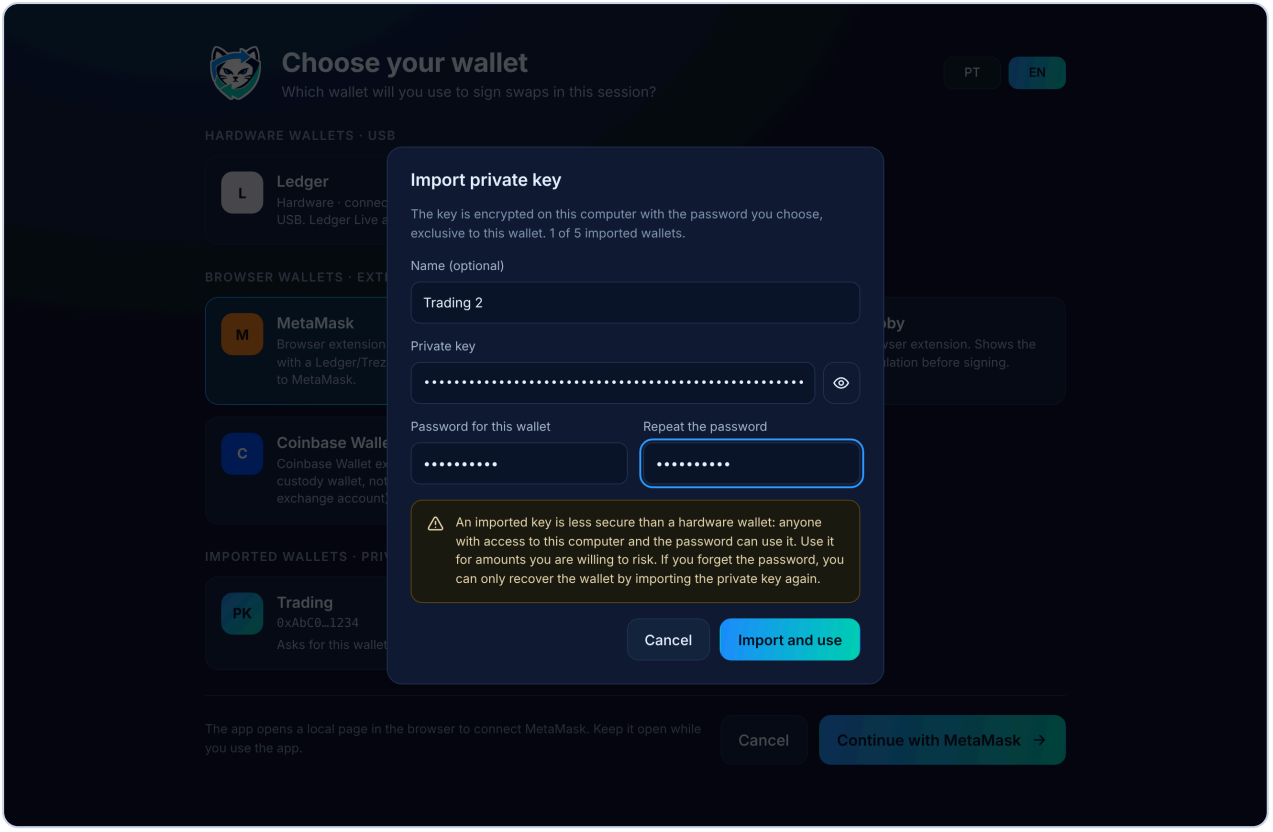


Figure 5. Importing a private key: name, key and a password exclusive to this wallet.

- When you choose an imported wallet, it asks for that wallet's password.
- It stays unlocked until you close the page, click “Lock”, or 30 minutes pass without use.
- Clearing the browser's site data deletes the encrypted keys from it: keep your private keys backed up elsewhere.

> An imported key is less secure than a hardware wallet. Use it for amounts you are willing to risk.

8 Making a swap

1. Swap: choose the token to sell and the token to receive, and the amount (or Max).
2. Options: MEV protection, maximum slippage, speed and the Uniswap versions (V2, V3, V4).
3. Click “Analyze pools”: each pool is quoted with your real amount.
4. Pick the pool (the best execution is preselected) and click Continue.
5. Review the summary: guaranteed minimum, pool, gas and send method.
6. Confirm in the wallet. The progress shows the approvals, the swap and the result.

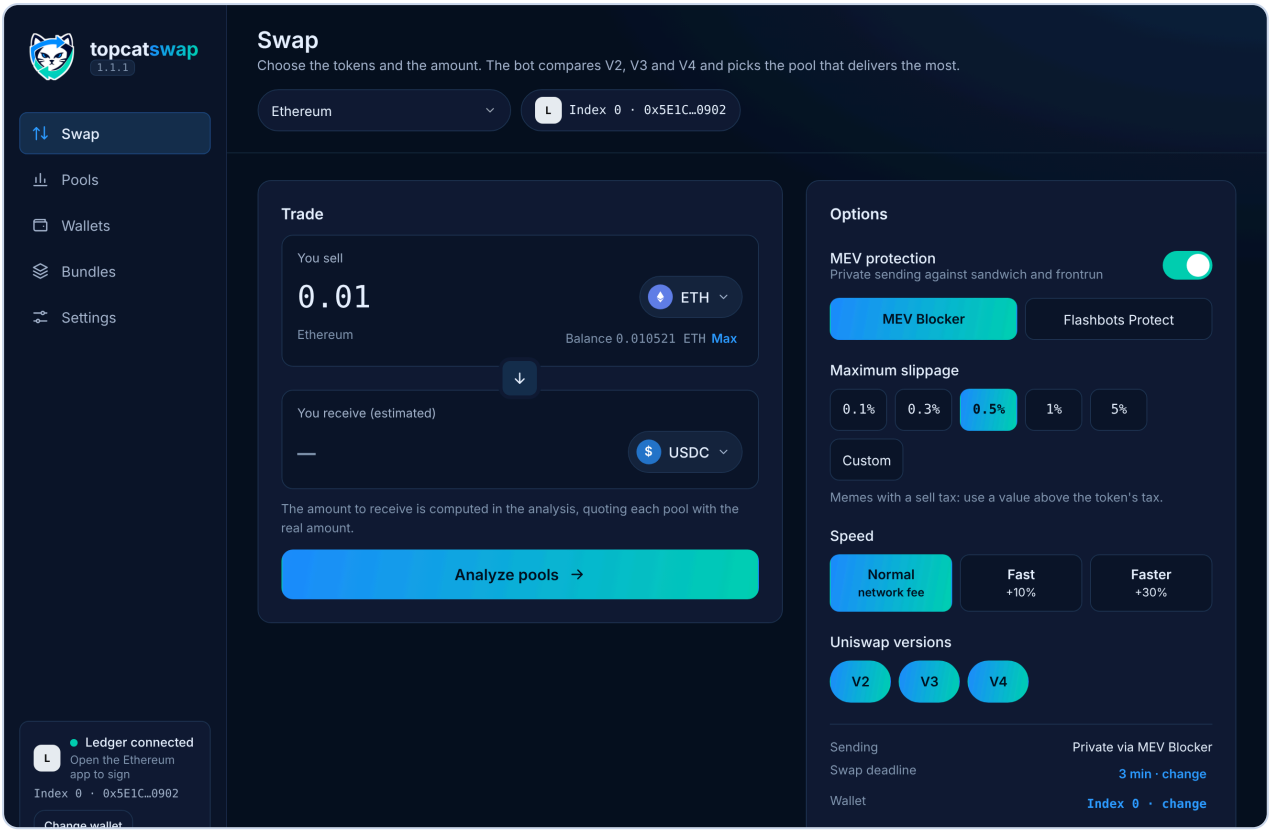


Figure 6. Swap screen with the options.

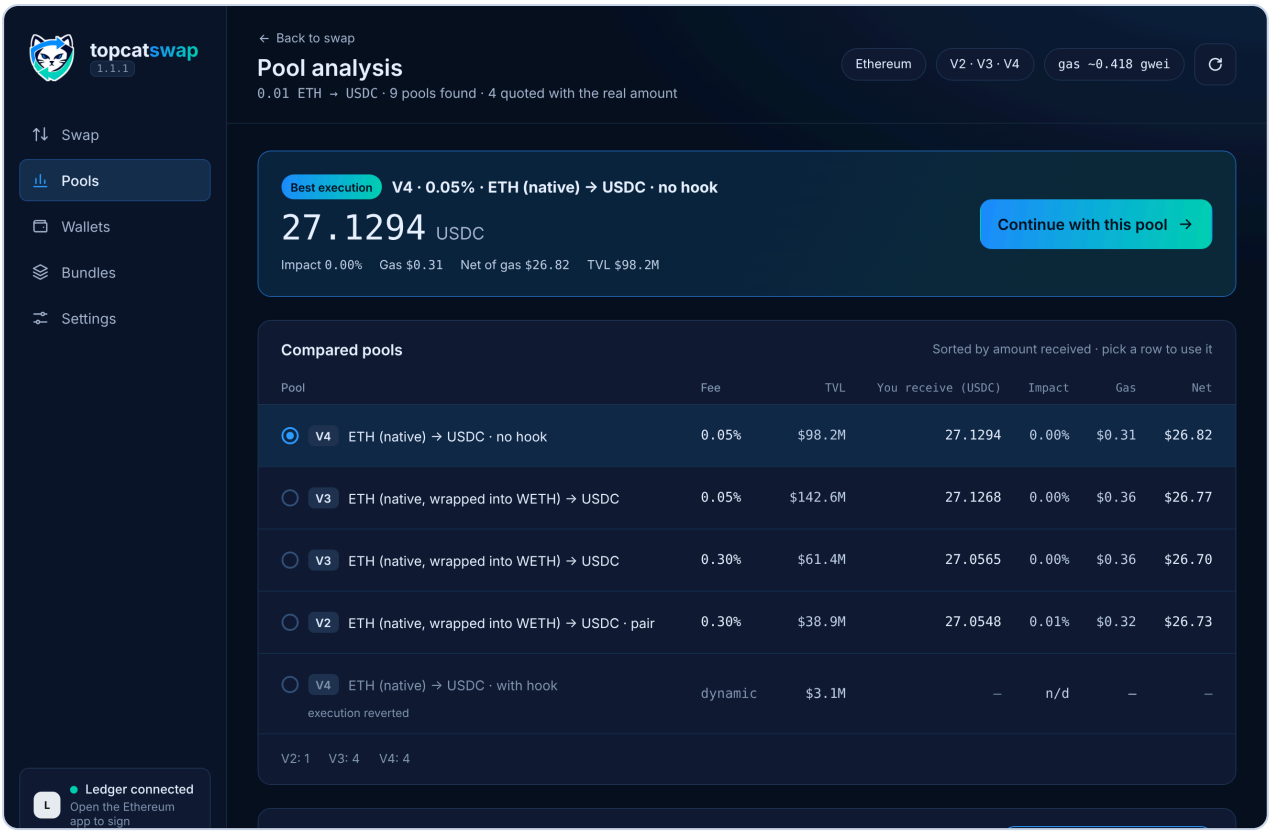


Figure 7. Pool analysis: best execution and the V2/V3/V4 comparison.

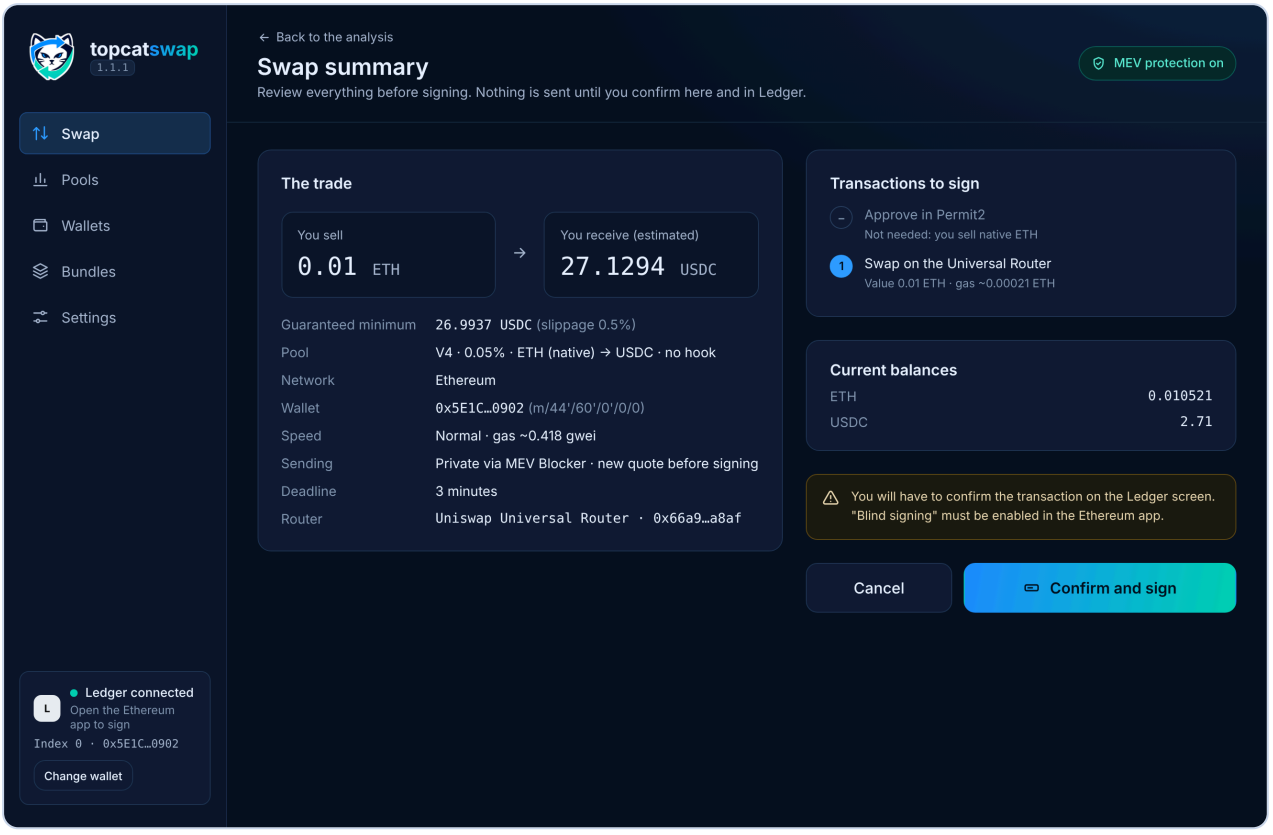


Figure 8. Swap summary before signing.

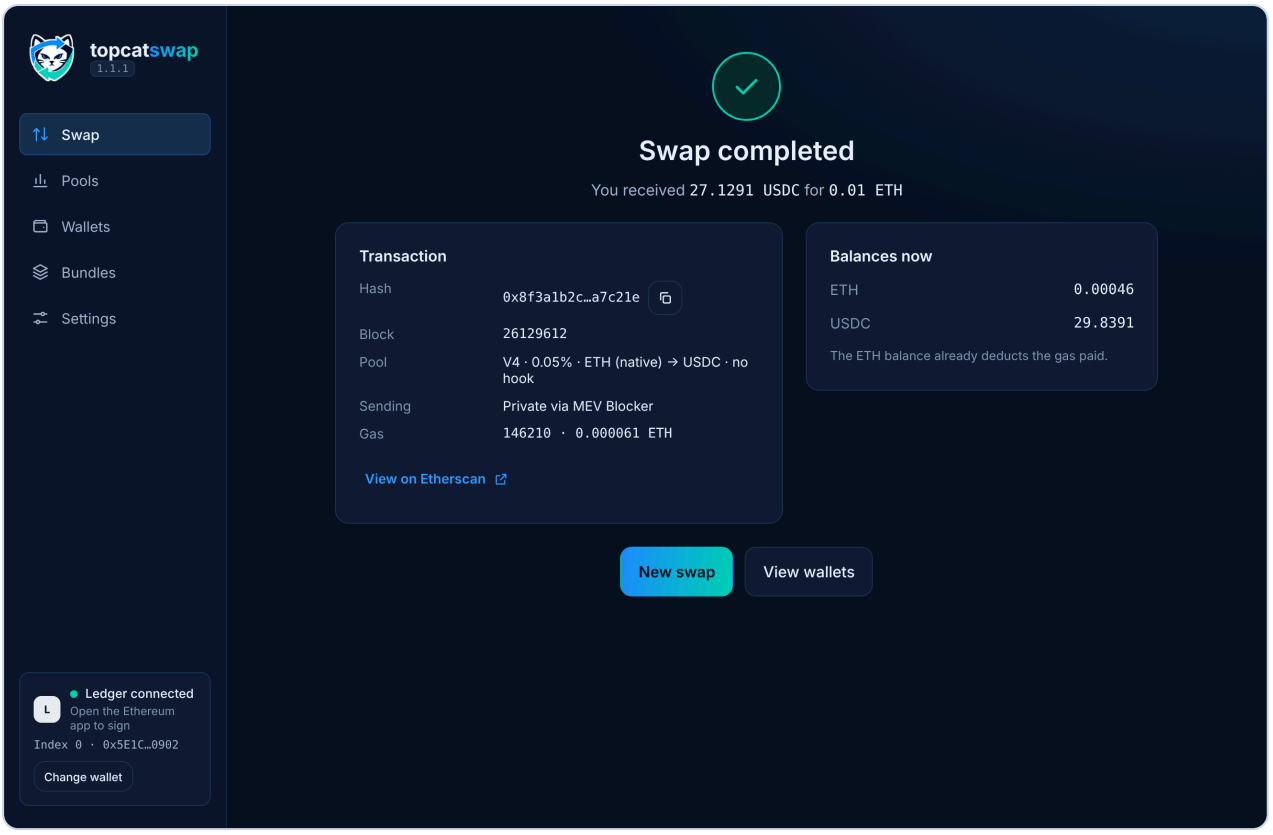


Figure 9. Swap completed.

> Do your first swap with a small amount. Tokens with a transfer tax: use V2 only and a slippage above the tax.

9 Bundles

The Bundles tab groups several transactions into a single block and sends them privately to the block builders instead of the public mempool.

Advantage	What it means
All or nothing	All transactions land in the same block, in order, or none does.
No gas if it fails	Builders simulate the bundle and skip it if anything fails: you pay nothing.
Private	Never in the public mempool: no sandwich, no frontrun.
Several builders	Ethereum: Flashbots, BuilderNet, Titan, beaverbuild. BSC: 48 Club, BlockRazor.

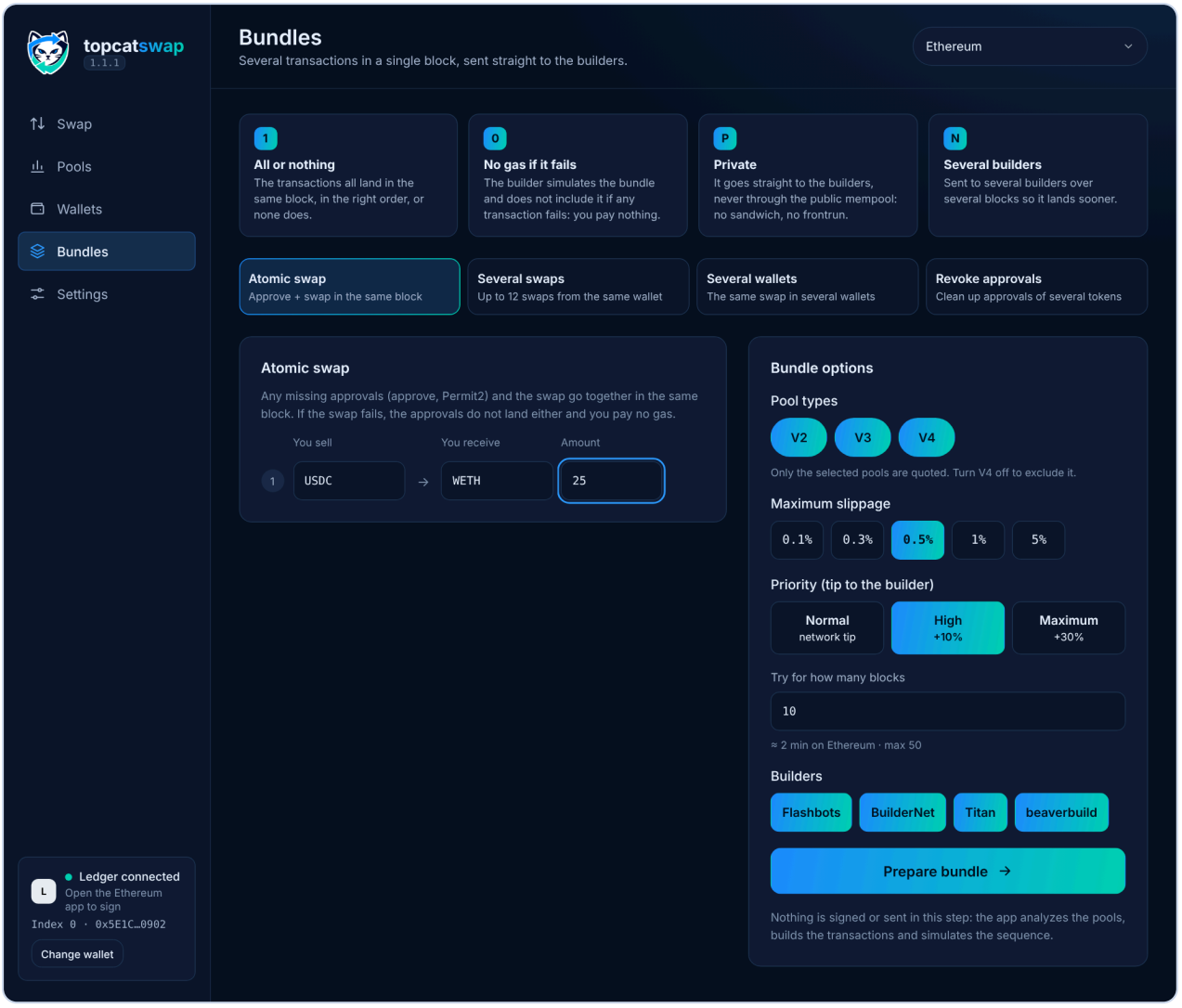


Figure 10. Bundles tab: types, pool types and options.

Types

- Atomic swap: missing approvals (approve, Permit2) and the swap in the same block.
- Several swaps: up to 12 swaps from the active wallet in one block.
- Several wallets: the same swap from several Ledger/Trezor accounts or imported wallets.
- Revoke approvals: removes approvals given to Permit2 and the Universal Router, in one bundle.

Options

- Pool types: choose V2, V3 and/or V4. Turn V4 off to exclude V4 pools.
- Maximum slippage.
- Priority: the tip to the builder (Normal, High, Maximum). A higher tip lands faster.
- Blocks: for how many blocks the bundle is tried (default 10 on Ethereum ≈ 2 min, 40 on BSC).
- Builders: where to send it.

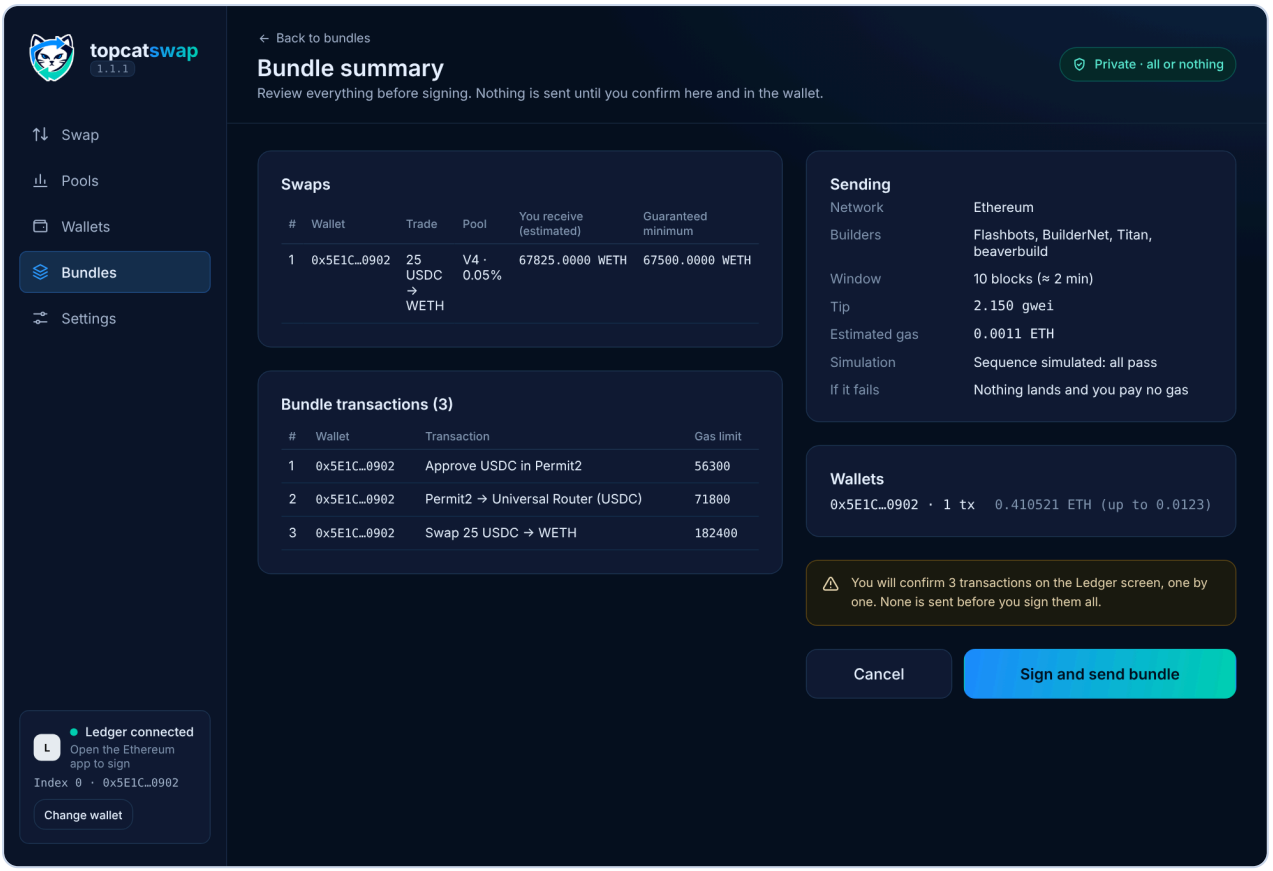


Figure 11. Bundle summary before signing.

On the web: the relay

Builders do not accept requests coming from websites. The web app sends bundles through a small relay that only forwards already-signed transactions to a fixed list of builders. Set its address in Settings › Connections › Bundle relay (see the Installation guide). The desktop app does not need it.

> If your RPC cannot simulate the sequence before signing, the summary says so and the bundle still goes out with a generous gas limit: builders simulate it anyway and skip it if it fails.

> Bundles work only on Ethereum and BSC, with a Ledger, Trezor or imported wallet.

10 MEV protection

With protection on, the swap goes to a private RPC and never enters the public mempool, where bots look for transactions to sandwich.

Network	Services
Ethereum	MEV Blocker, Flashbots Protect
BNB Smart Chain	Merkle, PancakeSwap, 48 Club
Arbitrum One	No public mempool: not needed

Slippage remains the final limit: it caps the worst price you accept.

11 Settings

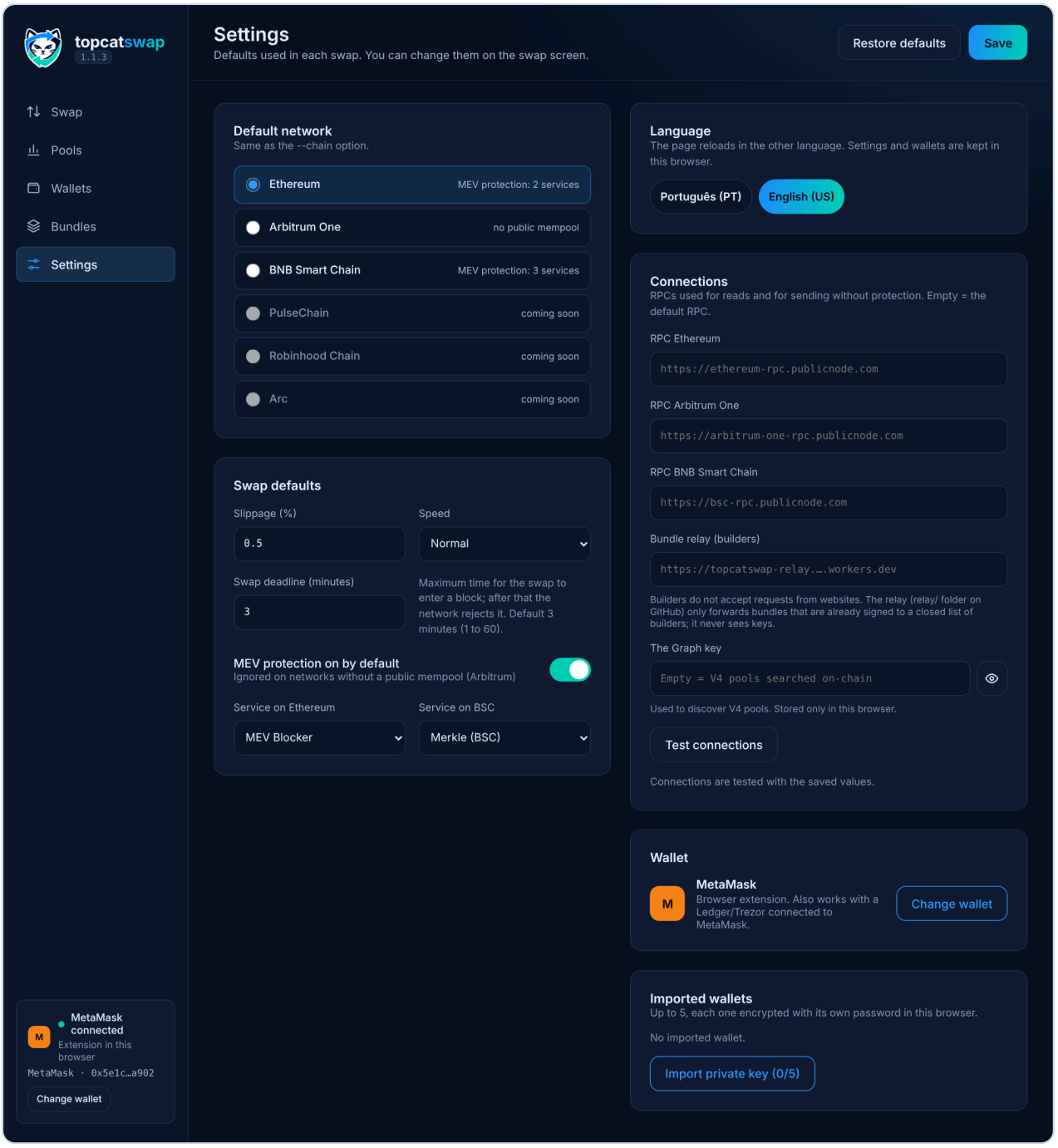


Figure 12. Settings on the web app, with the bundle relay field.

- Default network, slippage, speed, swap deadline and MEV protection.
- RPC per network (empty = default public RPC).
- Bundle relay (web only).
- The Graph key (optional): finds more V4 pools. Without it, standard V4 pools are searched on-chain.
- Language: English or Português.
- Imported wallets: use, lock or remove.

12 Security and verification

- No server, no accounts, no telemetry. Keys never leave the wallet or the browser.
- Every signed transaction is checked against the chosen wallet's address before sending.
- The site's pages load scripts with integrity hashes (SRI) and publish SHA256SUMS.txt: rebuild from the source code and compare.
- Source code and security policy: https://github.com/TOPCATMASTERINVESTOR/topcatswap

13 Troubleshooting

Problem	What to do
Ledger is “Unavailable”	Use Chrome, Edge or Brave on a computer. On Android use Bluetooth (Nano X, Stax, Flex).
No hardware wallets on iPhone/iPad	iOS gives websites no USB or Bluetooth. Use a wallet app (Open app) or an imported wallet.
“Open app” does nothing	The wallet app is not installed. Install it, or open the site in its in-app browser manually.
Extension not found	Install or enable the extension in this browser and reload the page.
Builders do not accept direct requests	Set the bundle relay in Settings › Connections.
Bundle not included	Nothing was executed or paid. Raise the priority or the blocks and try again.
Bundle simulation fails on a V4 pool	Turn V4 off under Pool types and prepare again.
Site in the wrong language	Tap PT or EN; the choice is saved in this browser.
Old version after an update	Reload the page (on phones, close the tab and open it again).